



Getting Started with Kali Linux

Verify / Write / Boot Lab menu, 16 steps

VERIFY THE DOWNLOAD

1	Lab directory Create and enter a working folder.	p. 2
2	Check tools and space Confirm GPG, SHA-256 and free disk.	p. 2
3	Download checksums Get the hash list and its signature.	p. 3
4	Import Kali key Add Kali's public signing key to GPG.	p. 4
5	Check fingerprint Confirm the key is the expected one.	p. 4
6	Verify signature Prove SHA256SUMS is signed by Kali.	p. 4
7	Find ISO hash Extract the official hash for the ISO.	p. 5
8	Check ISO hash Compare your ISO with the official hash.	p. 6

PREPARE THE USB

9	Identify devices Find the correct target before any write.	p. 6
10	Unmount USB Detach the partition before raw writing.	p. 6
11	Write ISO with dd Copy the image onto the whole USB device.	p. 7
12	Flush and verify Make sure all data is written.	p. 7

BOOT AND EXPLORE

13	Boot menu Boot from the USB and review the options.	p. 8
14	Live storage test See that Live changes are temporary.	p. 8
15	Edit boot options Remove quiet to see verbose boot output.	p. 9
16	Forensics mode Verify that storage is not auto-mounted.	p. 9

[VERIFY THE DOWNLOAD / STEP 1 OF 16](#)

Lab directory

Create and enter a working folder.

COMMAND

```
mkdir -p ~/lab2-kali  
  
cd ~/lab2-kali  
  
pwd
```

WHAT EACH PART DOES

<code>mkdir -p</code>	Create directory; no error if it exists.
<code>~</code>	Your home directory.
<code>pwd</code>	Print current directory.

[VERIFY THE DOWNLOAD / STEP 2 OF 16](#)

Check tools and space

Confirm GPG, SHA-256 and free disk.

COMMAND

```
gpg --version  
  
sha256sum --version  
  
df -h /
```

WHAT EACH PART DOES

<code>gpg --version</code>	GnuPG 2.4.9 installed.
<code>sha256sum</code>	GNU coreutils 9.10.
<code>df -h /</code>	Human-readable usage of root filesystem (/dev/sda1: 19G, 85% used).

OBSERVED OUTPUT Screenshot 1: gpg 2.4.9 (libgcrypt 1.12.2), sha256sum 9.10 (Debian 9.10-1). df -h / shows /dev/sda1 19G, 15G used, 2.7G available, 85%, mounted on /.

VERIFY THE DOWNLOAD / STEP 3 OF 16

Download checksums

Get the hash list and its signature.

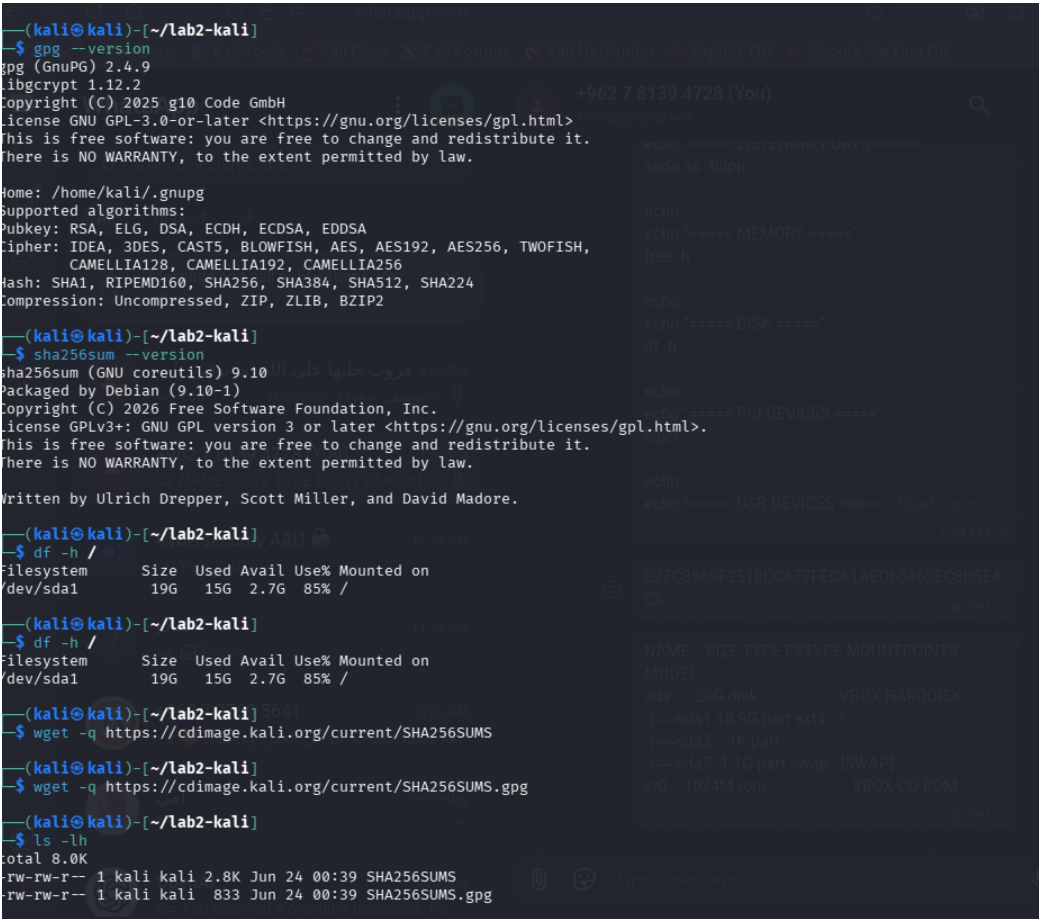
COMMAND

```
wget -q https://cdimage.kali.org/current/SHA256SUMS
wget -q https://cdimage.kali.org/current/SHA256SUMS.gpg
ls -lh
```

WHAT EACH PART DOES

-q	Quiet, no progress output.
SHA256SUMS	Official hash list (2.8K).
SHA256SUMS.gpg	Detached signature (833 bytes).

OBSERVED OUTPUT Screenshot 1: ls -lh lists SHA256SUMS (2.8K) and SHA256SUMS.gpg (833 bytes), owner kali, mode -rw-rw-r--, dated Jun 24 00:39.



Screenshot 1 - Tool versions, disk usage, checksum and signature downloads, file listing (Steps 2-3)

VERIFY THE DOWNLOAD / STEP 4 OF 16

Import Kali key

Add Kali's public signing key to GPG.

COMMAND

```
wget -q -O - https://archive.kali.org/archive-key.asc | gpg --import
```

WHAT EACH PART DOES

<code>-O -</code>	Send the download to stdout.
<code> </code>	Pipe it into the next command.
<code>gpg --import</code>	Store the key. ED65462EC8D5E4C5 imported.

OBSERVED OUTPUT Screenshot 2: key ED65462EC8D5E4C5 imported as 'Kali Linux Archive Automatic Signing Key (2025)'. Total processed: 1, imported: 1.

VERIFY THE DOWNLOAD / STEP 5 OF 16

Check fingerprint

Confirm the key is the expected one.

COMMAND

```
gpg --fingerprint 827C8569F2518CC677FECA1AED65462EC8D5E4C5
```

WHAT EACH PART DOES

<code>rsa4096</code>	4096-bit RSA key, expires 2028-04-17.
<code>[SC]</code>	Can sign and certify.
<code>[unknown]</code>	No local trust level set yet.

OBSERVED OUTPUT Screenshot 2: pub rsa4096 2025-04-17 [SC] [expires: 2028-04-17]. Fingerprint 827C 8569 F251 8CC6 77FE CA1A ED65 462E C8D5 E4C5. uid [unknown].

VERIFY THE DOWNLOAD / STEP 6 OF 16

Verify signature

Prove SHA256SUMS is signed by Kali.

WARNING Expected: Good signature. The 'key is not certified' warning is normal for a freshly imported key; trust comes from the fingerprint check.

COMMAND

```
gpg --verify SHA256SUMS.gpg SHA256SUMS
```

WHAT EACH PART DOES

<code>arg 1</code>	The signature file.
<code>arg 2</code>	The data file it covers.

OBSERVED OUTPUT Screenshot 2: signature made Wed 24 Jun 2026 12:39:32 AM EDT with RSA key 827C8569F2518CC677FECA1AED65462EC8D5E4C5. Good signature, plus the 'not certified with a trusted signature' warning.

VERIFY THE DOWNLOAD / STEP 7 OF 16

Find ISO hash

Extract the official hash for the ISO.

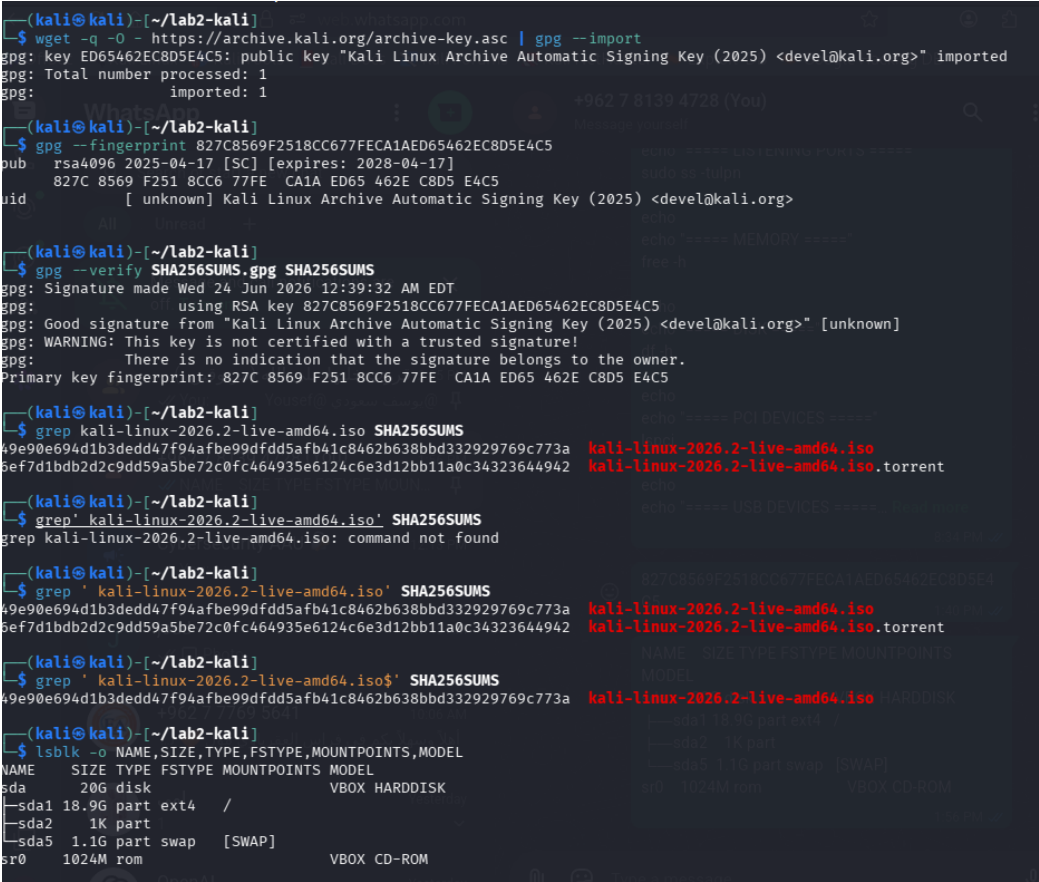
COMMAND

```
grep ' kali-linux-2026.2-live-amd64.iso$' SHA256SUMS
```

WHAT EACH PART DOES

' ... '	Quote the whole pattern. Quoting right next to grep caused 'command not found'.
\$	End of line; excludes the .torrent entry.
hash	49e90e69...769c773a

OBSERVED OUTPUT Screenshot 2: the unquoted grep matches the ISO and its .torrent line. The grep' typo gives 'command not found'. The leading-space pattern still matches both lines. Adding \$ leaves only the ISO line: 49e90e694d1b3dedd47f94afbe99dfdd5afb41c8462b638bbd332929769c773a.



Screenshot 2 - Key import, fingerprint, signature verification, grep ISO hash searches, lsblk (Steps 4-7 and 9)

VERIFY THE DOWNLOAD / STEP 8 OF 16

Check ISO hash

Compare your ISO with the official hash.

COMMAND

```
sha256sum kali-linux-2026.2-live-amd64.iso  
  
grep 'kali-linux-2026.2-live-amd64.iso$' SHA256SUMS | sha256sum -c
```

WHAT EACH PART DOES

sha256sum	Compute the local hash.
-c	Check mode; prints OK on match.

PREPARE THE USB / STEP 9 OF 16

Identify devices

Find the correct target before any write.

WARNING Never assume the USB is /dev/sdb. Check size and model first.

COMMAND

```
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS,MODEL
```

WHAT EACH PART DOES

sda	20G VBOX HARDDISK, system disk.
sda1 / sda5	ext4 root and swap partitions.
sr0	VBOX CD-ROM (rom).
sdb	Kingston USB 29G (from notes).

OBSERVED OUTPUT Screenshot 2: sda 20G disk VBOX HARDDISK; sda1 18.9G part ext4 on /; sda2 1K part; sda5 1.1G part swap [SWAP]; sr0 1024M rom VBOX CD-ROM. The Kingston USB (sdb) is recorded in the lab notes.

PREPARE THE USB / STEP 10 OF 16

Unmount USB

Detach the partition before raw writing.

COMMAND

```
sudo umount /dev/sdb1  
  
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS,MODEL
```

WHAT EACH PART DOES

sudo	Run as root.
umount	Detach the filesystem.
/dev/sdb1	First partition of the USB.

PREPARE THE USB / STEP 11 OF 16

Write ISO with dd

Copy the image onto the whole USB device.

WARNING DESTRUCTIVE. Everything on the target is erased. A wrong of= can wipe your system disk.

COMMAND

```
sudo dd if=~/lab2-kali/kali-linux-2026.2-live-amd64.iso of=/dev/sdb bs=4M status=progress conv=fsync
```

WHAT EACH PART DOES

if=	Input file (the ISO).
of=/dev/sdb	Output: whole disk, not sdb1.
bs=4M	4 MiB blocks.
status=progress	Show progress.
conv=fsync	Flush to device before exit.

PREPARE THE USB / STEP 12 OF 16

Flush and verify

Make sure all data is written.

COMMAND

```
sync  
  
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS,MODEL
```

WHAT EACH PART DOES

sync	Force buffered writes to storage.
lsblk	Re-inspect the USB layout.

BOOT AND EXPLORE / STEP 13 OF 16

Boot menu

Boot from the USB and review the options.

COMMAND

```
whoami
pwd
ls
```

WHAT EACH PART DOES

Live	Run without installing.
failsafe	Minimal drivers.
forensics mode	No automount, no swap.
Persistence	Keep changes on USB.
Installer	Install to disk.
Hardware / Memory	Detection and RAM test.

BOOT AND EXPLORE / STEP 14 OF 16

Live storage test

See that Live changes are temporary.

WARNING The course's bs=4M count=6144 writes 24 GiB. The 100 MiB version is safer. After a reboot, live-test.txt should be gone.

COMMAND

```
dd if=/dev/zero of=test.img bs=1M count=100
ls -lh test.img
rm test.img
echo "LAB 2 TEST" > ~/live-test.txt
cat ~/live-test.txt
```

WHAT EACH PART DOES

/dev/zero	Endless zero bytes.
bs=1M count=100	100 MiB total.
>	Redirect text into a file.

BOOT AND EXPLORE / STEP 15 OF 16

Edit boot options

Remove quiet to see verbose boot output.

COMMAND

```
Tab (Syslinux) or e (GRUB) on the Live entry

delete: quiet
```

WHAT EACH PART DOES

quiet	Hides boot messages.
removed	More output for troubleshooting.

BOOT AND EXPLORE / STEP 16 OF 16

Forensics mode

Verify that storage is not auto-mounted.

COMMAND

```
mount

lsblk

cat /etc/X11/Xsession.d/52kali_noautomount
```

WHAT EACH PART DOES

noswap	Do not use swap.
noautomount	Do not mount storage automatically.
mount / lsblk	Check for mounted internal disks.

SCREENSHOT INDEX

Screenshot	Description	Related steps
Screenshot 1	gpg and sha256sum versions, df -h /, wget -q downloads, ls -lh of SHA256SUMS and SHA256SUMS.gpg	2, 3
Screenshot 2	Key import, fingerprint, signature verification, grep ISO hash searches, lsblk	4, 5, 6, 7, 9